

	Infusion Pump Safety Requirements				
	Project:	PCA Infusion pump	Symbol:		Version: 1.1
	Author:		Status:		Date: 14.10.2021

Infusion Pump Safety Requirements

1 Equipment requirements

#	Requirement	Priority	Hazard traceability
RE.1	...		...
RE.2	...		...
RE.3	It's not possible to attach the line to the pump in a way it is out of downstream monitor sensor range		SR.D3
RE.5	...		...
RE.6	...		...

2 Hardware requirements

#	Requirement	Priority	Hazard traceability
RH.1	...		...
RH.2	...		...
RH.3	Downstream monitor rises alarm 'air-in-line' when air bubble larger than 50 µl is in line in the range of monitor sensor		SR.D1
RH.4	...		...
RH.5	'Downstream monitor error' alarm is raised when downstream monitor 'operation' signal is off		SR.D2
RH.6	...		...
RH.7	...		...

3 Software requirements

#	Requirement	Priority	Hazard traceability
RS.1			...
RS.2			...
RS.3	Alarm sound activated and error message on the screen is displayed on 'air-in-line' or 'Downstream monitor error' alarm signal		SR.C1
RS.4			...
RS.5	Pump is halted on 'air-in-line' or 'Downstream monitor error' alarm signal		SR.C2
RS.6			...
RS.7			...

4 Operation requirements

#	Requirement	Priority	Hazard traceability
RO.1			
RO.2	Compatible infusion set is used which when properly used protects the delivery path		SR.P1
RO.3			
RO.4	The infusion set is properly installed and the delivery path is sealed		SR.P2
RO.5			
RO.6	Air is expelled from line before infusion		SR.P3
RO.7			